

llmXive Automated Review of Macaron-A2UI: A Model for Generative UI in Personal Agents

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several specific quantitative claims that are not directly supported by the provided tables or text, creating a disconnect between the narrative and the evidence.

In Section 5.2 (“Main Results”), the text explicitly states: “Macaron-A2UI-Venti (GLM-5.1): 75.6 overall (w/o schema), surpassing GPT-5.4 (74.1) with full schema.” However, Table 1 (“Full-schema prompted baselines vs. schema-light Macaron models”) lists the “Avg.” score for GPT-5.4 as **3.54** and for Macaron-A2UI-Venti as **3.72**. The values 75.6 and 74.1 do not appear in the table. While it is plausible these are scaled scores (e.g., $3.72 * 20 = 74.4$, which is close but not exact to 75.6), the paper fails to define the scaling factor or explain the discrepancy. Without this clarification, the claim that the model “surpasses” the baseline by a specific margin is unverifiable based on the provided text.

Similarly, in Section 5.2, the text claims: “SFT improves Qwen-30B overall from 19.8 to 37.2; RL reaches 58.8.” These specific numbers are absent from Table 1 and the surrounding text. The table only lists the final results for the Macaron models and the frontier baselines (GPT-5.4, etc.). The reader cannot verify the “19.8” base score or the “58.8” RL score without an external reference or a missing table. This makes the claim of “large margin” improvement unsupported by the visible evidence.

In Section 4.3 (“Validation and Repair”), the text states: “Final renderability: 99.2% (only 85 failures after 3 attempts).” A simple calculation on the total corpus size of 14,245 samples (from Table 2) shows that 99.2% success corresponds to approximately 115 failures ($14,245 * 0.008 \approx 114$). If there were only 85 failures, the success rate would be $14,160/14,245 \approx 99.4\%$. The numbers 99.2% and 85 are mathematically inconsistent with the total sample count provided in the same section.

These issues suggest that the authors may be referencing a different version of the results table or have made arithmetic errors in the text. The claims regarding performance superiority and dataset quality are currently too strong given the mismatch between the narrative numbers and the tabular data.

Action items.

- **[science]** Section 5.2 (Main Results) claims Macaron-A2UI-Venti (75.6) surpasses GPT-5.4 (74.1). However, Table 1 lists GPT-5.4 Avg as 3.54 and Macaron-A2UI-Venti as 3.72. The text implies a direct comparison of the ‘75.6’ and ‘74.1’ scores, but these values do not appear in the table. Clarify if these are scaled percentages (e.g., $3.72 * 20$) or if the table values are incorrect. The claim is currently unsupported by the provided data.
- **[writing]** Section 4.3 states ‘Final renderability: 99.2% (only 85 failures after 3 attempts).’ The math (14,245 total samples) implies ~115 failures for 99.2% success, not 85. If 85 is the correct failure count, the percentage should be ~99.4%. If 99.2% is correct, the failure count is wrong. Verify the arithmetic in Section 4.3.
- **[science]** Section 5.2 states ‘SFT improves Qwen-30B overall from 19.8 to 37.2; RL reaches 58.8.’ These specific numbers (19.8, 37.2, 58.8) are not present in Table 1 or the main text body. They appear to be derived from a different metric or a missing table. Cite the specific table or figure where these baseline scores are reported to support the claim.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 effectively illustrates the paper’s core concept by showing a chat interface where a user’s abstract goal is distilled into a concrete, actionable reminder card. The visual hierarchy is clear, the text is legible, and the figure aligns perfectly with its caption.

Figure 2

Figure 2 effectively demonstrates the model’s capability to distill a dialogue into a structured UI component. The chat context clearly establishes the requirements (school nights, neighbor lead), and the generated ‘Childcare plan’ card accurately reflects these details with actionable options, fully supporting the caption’s claim.

Figure 3

The figure illustrates a decision card interface effectively, but the underlying dialogue text includes a significant factual error regarding COVID-19 mortality rates that contradicts the ‘supportive counseling’ context.

Figure 4

Figure 4 effectively demonstrates the paper’s claim by showing a chat interface where a booking request is resolved into a compact, structured confirmation card. The visual elements, including the ‘Book taxi’ form with specific fields like ‘Leave by 08:45’ and a ‘Confirm’ button, are clearly legible and align perfectly with the caption’s description of a transaction state verification surface.

Figure 5

Figure 5 effectively illustrates the paper’s claim by contrasting a verbose plain-text dialogue with a streamlined, structured interface. The visual layout is clear, with distinct sections for input methods and output results, and the accompanying icons and labels are legible and well-defined.

Figure 6

Figure 6 is a clear and well-structured pipeline diagram that effectively visualizes the A2UI corpus construction process. The flow is logical, the steps are clearly labeled with icons, and the final statistics are legible. The caption accurately describes the workflow shown in the image.

Figure 7

Figure 7 is a clear and well-constructed horizontal bar chart that effectively visualizes the top-10 component frequencies. The axes are labeled with units, data values and percentages are explicitly provided on the bars, and the caption accurately describes the content.

Figure 8

Figure 8 presents a clear ablation study with distinct panels, but the legend is overly cluttered and fails to explicitly map the ‘untuned frontier references’ mentioned in the caption to specific bars, creating ambiguity in the comparison.

Figure 9

The figure effectively visualizes the reward curves for two model sizes, but the caption is insufficiently descriptive and the lack of error bars obscures the reliability of the reported trends.

Action items.

- **[science]** Figure 3: The dialogue contains a hallucinated and potentially harmful medical claim (‘survivor rate from COVID-19 infections is around 99%’) that is factually inaccurate and contextually inappropriate for a counseling exchange, undermining the figure’s claim of a ‘supportive’ interaction.
- **[science]** Figure 8: The legend lists ‘GPT-4o mini’ and ‘DeepSeek V3.1’ as solid red and green bars, but the caption describes the left-side bars as ‘untuned, SFT, and SFT+RL models’ plus ‘untuned frontier references’. The legend fails to distinguish which specific bars correspond to the ‘untuned frontier references’ versus the ‘untuned’ Macaron models, making the ablation comparison ambiguous.
- **[writing]** Figure 8: The legend is cluttered and difficult to parse, listing 12 distinct entries with varying color/pattern combinations (solid, hatched) that are not clearly mapped to the specific ‘w/o schema’ vs ‘w/ schema’ groups in the legend text itself, forcing the reader to guess which legend items apply to which panel section.
- **[writing]** Figure 9: The caption ‘Total reward’ is too brief to explain the plot’s context (e.g., training vs. evaluation, specific task) or the meaning of the ‘235B’ and ‘30B’ labels.
- **[science]** Figure 9: The plot displays jagged, noisy lines without error bars or shaded confidence intervals, making it difficult to assess the statistical significance of the performance difference between the 235B and 30B models.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on domain-specific acronyms and proprietary protocol terminology that significantly raises the barrier to entry for non-specialist readers. The most critical issue is the introduction of the core concept, “A2UI,” in the title and first paragraph without ever explicitly defining what the acronym stands for or what the protocol entails in plain language. This pattern repeats throughout the text.

In Section 3, the four message types (`surfaceUpdate`, `dataModelUpdate`, etc.) are listed as if they are common knowledge, yet they are specific to this paper’s proposed protocol. Similarly, in Section 5.1, the training methodology relies on “LoRA-based SFT” and “GRPO” without expanding these acronyms or explaining the underlying mechanisms in accessible terms. “GRPO” is particularly opaque; while it likely refers to Group Relative Policy Optimization, the paper does not state this, leaving a general AI researcher or HCI specialist guessing.

The term “schema-light” is used repeatedly (Introduction, Section 5.1) to describe the training approach. While the context suggests it means “without explicit schema hints,” the term itself is jargon that should be defined or replaced with a clearer phrase like “training without explicit schema constraints.” Additionally, “VLM” is used in Section 5.2 to describe the visual judge without expansion.

Finally, phrases like “heterogeneous dialogue sources” (Section 4) and “component-targeted augmented samples” (Section 4.1) add unnecessary density. Replacing “heterogeneous” with “diverse” and clarifying “component-targeted” would improve flow without losing precision. The paper assumes a level of familiarity with specific RLHF techniques and UI protocol design that excludes a broader audience.

Action items.

- **[writing]** The manuscript relies heavily on domain-specific acronyms and proprietary protocol terminology that significantly raises the barrier to entry for non-specialist readers. The most critical issue is the introduction of the core concept, “A2UI,” in the title and first paragraph without ever explicitly defining what the acronym stands for or what the protocol entails in plain language. This pattern repeats throughout the text. In Section 3, the four message types (`surfaceUpdate`, `dataModelUpdate`, etc

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent narrative regarding the construction of the A2UI corpus and the benchmark, but there are significant logical inconsistencies between the reported numerical results and the textual claims, as well as a gap in the causal mechanism of the training algorithm.

First, the central claim in the Introduction and Conclusion—that the best model (Macaron-A2UI-Venti) achieves a score of 75.6 and surpasses the strongest baseline (GPT-5.4 at 74.1)—is not supported by the data presented in Table 1 (Section 5.2). Table 1 explicitly lists the “Avg.” scores as 3.72 for Macaron-A2UI-Venti and 3.54 for GPT-5.4. While 3.72 is indeed higher than 3.54, the specific values 75.6 and 74.1 appear nowhere in the table. The text implies these are the direct scores, yet they differ by a factor of roughly 20 from the table values. Without a clear definition of how the “overall” score (75.6) is derived from the component scores (L1-L3, V1-V3) or the table averages, the conclusion that the model “surpasses” the baseline by a specific margin is logically ungrounded. The authors must either update the table to reflect the 75.6/74.1 scale or explicitly state the normalization formula used to convert the 1-5 scale to the 0-100 scale.

Second, there is a logical disconnect in the description of the GRPO training dynamics. The reward function is defined as $R = \mathbf{1}[\text{pass}] \cdot (\lambda_1 S_{L1} + \lambda_2 S_{L2} + \lambda_3 S_{L3})$. The term $\mathbf{1}[\text{pass}]$ acts as a gate, likely dependent on L1 (protocol validity). If a generated response fails the L1 check, the total reward R becomes zero. Logically, this means the model receives no gradient signal to improve L2 (Task Construction) or L3 (User Experience) for any sample that fails L1. However, the results section states that “L2/L3 improve gradually” and that “L3 is hardest,” implying that the model learns these complex skills even while L1 is still being mastered. If the reward is strictly zero for L1 failures, the model cannot learn L2/L3 until L1 is perfect, which contradicts the observed gradual improvement curves. The authors need to clarify if the “pass” condition is soft, if L2/L3 rewards are calculated independently of the L1 gate, or if the mechanism allows for partial credit.

Finally, the claim that the model internalizes the schema “without long schema prompts” relies on the comparison between the “w/o schema” Macaron models and the “w/ schema” baselines. While the results show Macaron outperforming the baselines, the logical leap that the model has “internalized” the schema rather than simply learning to mimic the output format via SFT is not fully supported. The paper does not provide an ablation study showing that the model fails when the A2UI protocol is slightly perturbed or when asked to generate a *new* component type not seen in training. Without this, the conclusion that the capability is “internalized” rather than “memorized” remains an assumption.

Action items.

- **[science]** The claim that Macaron-A2UI-Venti (75.6) surpasses GPT-5.4 (74.1) contradicts Table 1, which reports averages of 3.72 and 3.54. The text implies these are direct scores, but the values differ by a factor of ~20. Clarify the scaling formula or update the table to match the 0-100 scale used in the text.
- **[science]** The reward function $R = \mathbf{1}[\text{pass}] \cdot (\text{weighted sum})$ implies zero reward for L1 failures, preventing L2/L3 learning. Yet results show gradual L2/L3 improvement. Explain how the model learns L2/L3 when L1 is imperfect, or clarify if the ‘pass’ gate is soft/partial.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several strong claims regarding the efficiency and generalizability of the Macaron-A2UI model that appear to overreach the provided evidence.

First, the central claim in the Abstract and Conclusion—that the model “surpasses the strongest full-schema frontier baseline”—is potentially misleading due to a mismatch in model scale. Table 1 compares Macaron-A2UI-Venti (based on a 235B parameter model) against baselines like GPT-5.4 (implied 30B scale in the text: “GPT-5.4... 30B”). Attributing the performance gain (75.6 vs 74.1) primarily to the “schema-light” training strategy rather than the massive difference in parameter count (235B vs 30B) is an over-interpretation. The paper fails to isolate the variable of “schema prompting” from “model capacity.” A fairer claim would be that the model performs well *relative to its size* or that it matches smaller models with full schema, rather than surpassing them without qualification.

Second, the definition of “schema-light” is inconsistent. The Introduction and Conclusion emphasize that the model works “without explicit schema hints.” However, the System Prompt provided in Appendix A2UI Prompts (Section “System prompt w/o A2UI schema”) explicitly enumerates the allowed component names (e.g., “Button, Card, Column...”) and message types. While this is less detailed than a full JSON schema, it is still a rigid structural constraint provided at inference time. Claiming the model internalizes the protocol “without schema hints” ignores that the vocabulary and message structure are hard-coded into the prompt. The authors should refine this terminology to “schema-agnostic” or “vocabulary-constrained” rather than implying a complete absence of structural guidance.

Finally, the conclusion generalizes that “Generative UI capability... can be learned and internalized” as a universal finding. This extrapolates beyond the specific A2UI v0.8 protocol used in the study. The training data and reward functions are tightly coupled to the specific component catalog (23 types) and message types defined in A2UI. There is no evidence presented that this “internalization” transfers to other UI frameworks (e.g., raw HTML/CSS, Flutter, or React) without retraining. The claim should be restricted to the specific protocol domain studied.

Action items.

- **[science]** The claim that the model ‘surpasses the strongest full-schema frontier baseline’ (Abstract, Conclusion) relies on a comparison between a 235B parameter model (Macaron-Venti) and a 30B parameter baseline (GPT-5.4). This conflates model scale with architectural efficiency. The paper must clarify if the ‘surpassing’ claim holds when comparing models of similar scale or if the advantage is solely due to the larger parameter count.
- **[writing]** The introduction states the model achieves 75.6 overall ‘without explicit schema hints,’ yet the system prompt in Appendix A2UI Prompts explicitly lists 23 allowed component

names and message types. This constitutes a form of schema constraint. The authors must distinguish between ‘full schema’ (detailed field definitions) and ‘component vocabulary’ to avoid over-claiming the ‘schema-light’ nature of the inference.

- **[writing]** The conclusion asserts that ‘Generative UI capability... can be learned and internalized’ based on results from a single protocol (A2UI v0.8). This generalization overreaches the data scope. The paper should temper claims about generalizability to other UI protocols or frameworks, as the training is tightly coupled to the specific A2UI message types and component catalog.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript presents a novel approach to Generative UI but requires clarification on several safety and ethical dimensions before publication.

First, regarding **data privacy and consent**, the authors construct a corpus from heterogeneous sources including MultiWOZ, ESConv (Emotional Support Conversation), and AnnoMI (Motivational Interviewing). While these datasets are publicly available, the paper (Section 4) does not explicitly state whether the original data licenses permit the creation of a new, derived benchmark (A2UI-Bench) or if the transformation process involved any re-identification risks. Specifically, the ESConv dataset contains sensitive mental health dialogues. The authors must include a dedicated “Ethics Statement” or “Data Usage” section confirming that the derived dataset adheres to the original consent terms and that no personally identifiable information (PII) was inadvertently exposed or amplified during the augmentation process.

Second, the **evaluation metrics** introduce potential safety risks through arbitrary thresholds. In Section 5.2 (Metrics), the L3 (User Experience) evaluation includes a “Strict deduction” rule: “More than 4 independent interactive components or more than 8 options displayed at once” results in a score of 1. This threshold appears arbitrary and is not grounded in established Human-Computer Interaction (HCI) literature or accessibility guidelines (such as WCAG). Enforcing such a rigid limit could penalize models that correctly identify a need for complex interaction in high-stakes scenarios (e.g., medical triage or financial planning) and could inadvertently bias the model against providing necessary information density. The authors should justify this threshold with empirical evidence or cite relevant cognitive load studies.

Finally, there is a **dual-use and bias concern** regarding the generated UI components. The system prompts (Appendix A.1) explicitly list icon names including “emotion-unhappy,” “anguished-face,” and “disappointed-face.” In the context of the ESConv and AnnoMI datasets, the generation of UIs containing negative emotional icons could be psychologically harmful or trigger distress in vulnerable users. The paper currently lacks a discussion on how the model is constrained to prevent the generation of such potentially harmful visual elements in sensitive

conversational contexts. A safety mitigation strategy or a discussion on the ethical implications of “emotional UI” is required.

Action items.

- **[writing]** The paper lacks an explicit statement regarding IRB approval or ethical review for the use of human dialogue datasets (MultiWOZ, ESConv, AnnoMI). While these are public, the transformation into a new benchmark and the potential for generating sensitive UI interactions in emotional support contexts requires a dedicated ethics statement confirming compliance with original data licenses and user consent.
- **[writing]** The ‘L3 Cognitive Load’ metric (Section 5.2) defines a strict deduction for ‘more than 4 independent interactive components’ as a failure. This arbitrary threshold lacks empirical justification regarding human factors or accessibility standards (e.g., WCAG). The paper should cite relevant HCI literature or user studies to validate this specific cutoff to avoid bias against complex but necessary interactions.
- **[writing]** The system prompt for the ‘w/o schema’ model (Appendix A.1) lists specific icon names (e.g., ‘emotion-unhappy’, ‘anguished-face’). The paper does not address the safety implications of generating UIs with negative emotional icons in sensitive contexts (e.g., ESConv). A discussion on mitigating potential harm or bias in icon selection for mental health scenarios is needed.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence supporting the central claim—that the Macaron-A2UI model can internalize Generative UI capabilities without schema hints and outperform frontier baselines—is currently insufficient due to statistical and methodological gaps.

First, the benchmark size is a significant limitation. The A2UI-Bench contains only 300 tasks (Section 5.1). While the training corpus is large (14,245 samples), the evaluation set is small. The paper claims the Macaron-A2UI-Venti model (75.6) surpasses the GPT-5.4 baseline (74.1) in Table 1 (Section 6.2). Without reported standard deviations, confidence intervals, or a statistical significance test (e.g., paired t-test or bootstrap), it is impossible to determine if this 1.5-point difference is a genuine improvement or statistical noise. Given the small N, the variance could easily exceed the observed effect size.

Second, the evaluation methodology relies entirely on automated judges (LLMs for L1-L3 and VLMs for V1-V3). The paper defines “User Experience” (L3) and “Task Alignment” (V2) but provides no evidence that these automated metrics correlate with human judgment. There is no mention of a human evaluation study, nor are inter-rater reliability metrics (e.g., Cohen’s Kappa) provided for the judges themselves. If the judges are biased or inconsistent, the reported scores

(e.g., 4.67 for L1) are not scientifically valid measures of quality. The “VLM judge” prompt (Appendix) is provided, but its calibration against human ground truth is absent.

Third, the training pipeline introduces a potential circularity risk. The GRPO reward function (Section 6.1) is explicitly defined as a weighted sum of the L1, L2, and L3 scores generated by the same LLM judges used for evaluation. This creates a high risk of reward hacking, where the model optimizes for the specific linguistic patterns or biases of the judge prompts rather than genuine UI utility. The paper does not discuss how they mitigated this, such as using a separate, held-out judge model or human-in-the-loop validation during RL.

Finally, the “schema-light” claim is supported by comparing the model against baselines that *use* schema prompts. However, the paper does not provide a control experiment where the Macaron model is tested *with* schema prompts to see if it could perform even better, nor does it analyze the failure modes where the lack of schema leads to protocol violations (L1) that the RL failed to correct. The evidence suggests the model works, but the robustness of the “internalization” claim is weakened by the lack of statistical rigor and independent validation of the evaluation metrics.

Action items.

- **[science]** The benchmark size (300 tasks) is small for robust statistical claims. Report confidence intervals or standard errors for the reported mean scores (e.g., 75.6 vs 74.1) to determine if the ‘surpassing’ claim is statistically significant or within noise.
- **[science]** The evaluation relies heavily on LLM and VLM judges (L1-L3, V1-V3). The paper lacks a human evaluation study or inter-rater reliability metrics (e.g., Cohen’s Kappa) to validate that these automated judges correlate with human perception of ‘User Experience’ (L3).
- **[science]** The training reward function (R) is a weighted sum of L1, L2, and L3 scores. Since the judges are the same models used to generate the reward, there is a high risk of reward hacking or overfitting to the specific prompt biases of the judge models rather than genuine UI quality.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The manuscript presents a comprehensive evaluation framework for Generative UI, utilizing a multi-dimensional scoring system (L1-L3, V1-V3). However, the statistical rigor of the reported results requires significant strengthening to support the paper’s central claims.

First, **Table 1** (Main Results) presents point estimates (e.g., 75.6 vs. 74.1) without any measure of variance (standard deviation, standard error) or confidence intervals. In machine learning evaluations, especially with LLM judges, scores can vary significantly based on prompt

sensitivity or random seeds. Without error bars or confidence intervals, it is impossible to determine if the observed difference between Macaron-A2UI-Venti and the GPT-5.4 baseline is statistically significant or within the noise margin of the evaluation protocol. The claim that the model “surpasses” the baseline is currently unsupported by statistical evidence.

Second, the **evaluation methodology** relies on LLM and VLM judges. The paper does not report inter-rater reliability (e.g., Cohen’s Kappa or Krippendorff’s Alpha) for these automated judges. Given that the metrics are subjective (e.g., “Cognitive Load,” “Conversational Naturalness”), establishing the consistency of the judges is a prerequisite for the validity of the scores. If the judges are inconsistent, the reported means are unreliable.

Third, the **sample size and unit of analysis** are ambiguous in the statistical context. While the benchmark contains 300 tasks, the text mentions “300 tasks, including negative cases.” It is unclear if the reported scores are averages over 300 independent trials or if multiple turns per dialogue were averaged. The effective sample size (N) for the statistical tests must be explicitly stated. If the unit of analysis is the dialogue turn rather than the task, the independence assumption for standard parametric tests (like t-tests) may be violated due to intra-dialogue correlation.

Finally, the **reward curves** in Figure 4 show training dynamics but lack statistical aggregation. If these curves represent a single run, they do not demonstrate the stability of the RL training. Reporting the mean and standard deviation of the final reward over multiple seeds (e.g., 3-5 runs) is standard practice to rule out lucky initialization or stochasticity in the reward signal.

To address these issues, the authors should re-run the evaluation to include variance metrics, perform significance testing (e.g., paired t-tests or Wilcoxon signed-rank tests) on the benchmark results, and report inter-rater reliability for the automated judges.

Action items.

- **[science]** Report confidence intervals or standard errors for all mean scores in Table 1 (e.g., 75.6, 74.1) to quantify uncertainty, as single-point estimates are insufficient for statistical comparison.
- **[science]** Clarify the statistical significance of the performance gap between Macaron-A2UI-Venti and GPT-5.4; a t-test or non-parametric equivalent is required to support the claim of ‘surpassing’ the baseline.
- **[science]** Define the unit of analysis for the evaluation metrics (e.g., per-turn vs. per-dialogue) and report the effective sample size (N) for each metric to ensure reproducibility of the statistical tests.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a clear and generally well-structured argument for the Macaron-A2UI model. The introduction effectively sets the stage, and the problem formulation is concise. However, the writing quality is significantly impacted by structural redundancy and minor ambiguities in data reporting.

The most critical issue is the duplication of content between the main body and the Appendix. Sections titled “A2UI Prompts” and “A2UI Rendering Implementation” appear in the main text (e000) and are repeated almost verbatim in the Appendix (e001). This repetition breaks the logical flow of the paper, making it feel disjointed and bloated. Standard academic practice dictates that detailed prompt templates and implementation specifics belong in the Appendix, with only high-level summaries in the main text. The authors should consolidate these sections to ensure a smooth reading experience.

Additionally, there are minor issues with clarity in the data presentation. In Section “A2UI Corpus Construction,” the summary sentence following Table 1 is slightly ambiguous regarding the composition of the total sample count. While the numbers are present in the table, the prose could be tightened to explicitly define the breakdown of UI-turns versus text-only turns to avoid any potential confusion for the reader.

Finally, the Conclusion could be strengthened by replacing vague qualitative descriptors like “slightly surpasses” with precise quantitative comparisons (e.g., “surpasses by 1.5 points”) when referencing the benchmark results. This would enhance the precision and authority of the closing argument. Addressing these structural and stylistic points will significantly improve the overall readability and professionalism of the paper.

Action items.

- **[writing]** The manuscript contains significant structural duplication between the main body and the Appendix. Specifically, Section ‘A2UI Prompts’ and ‘A2UI Rendering Implementation’ appear in both the main text (e000) and the Appendix (e001) with nearly identical content. This redundancy disrupts the narrative flow and should be resolved by moving detailed prompt templates and implementation specs entirely to the Appendix, referencing them briefly in the main text.
- **[writing]** In Section ‘A2UI Corpus Construction’ (e000), the text states ‘Total: 14,245 samples’ immediately after Table 1, but the table’s ‘Total’ row lists ‘14,245’ under ‘Samples’ while the ‘UI / Text’ column sums to 14,245 as well. The sentence structure is slightly ambiguous regarding whether the total includes the augmented samples or just the base. Clarify the sentence to explicitly state: ‘The final corpus comprises 14,245 samples, consisting of 10,210 UI-turns and 4,035 text-only turns.’
- **[writing]** In the ‘Conclusion’ (e001), the phrase ‘slightly surpasses the strongest full-prompt frontier baseline’ is vague. The preceding sentence mentions specific scores (75.6 vs 74.1). Replace ‘slightly surpasses’ with ‘surpasses by 1.5 points’ or similar specific phrasing to improve precision and impact.